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Faculty *of Mathematics and Physics*

Department of Physics

# Successive Over-Relaxation for Boundary Value Problems

**5. Assignment for Model Analysis 2, 2025/26**

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Ljubljana, March 2026

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# 1 Introduction

This week we finally move on from shooting methods to a more general class of methods for solving boundary value problems: relaxation methods. Specifically we will be taking a look at the Successive Over-Relaxation (SOR) [1] method, which is an iterative method for solving linear systems of equations that arise from discretizing partial differential equations (PDEs). The SOR method is an extension of the Gauss-Seidel method, which itself is an improvement over the Jacobi method. The key idea behind SOR is to introduce a relaxation factor that can accelerate the convergence of the iterative process. That is to say, that instead of simply using the new values computed in the current iteration, we intentionally "over-relax" the update by a factor $\omega$ (the relaxation factor), which can be tuned to achieve faster convergence.

The relaxation factor $\omega$ in bounded between 0 and 2, where $\omega = 1$ corresponds to the Gauss-Seidel method, $\omega < 1$ corresponds to under-relaxation, and $\omega > 1$ corresponds to over-relaxation. $\omega = 2$ corresponds to the limit of stability, and values of $\omega \geq 2$ will lead to divergence. The optimal choice of $\omega$ depends on the specific problem being solved, and finding it can significantly reduce the number of iterations required for convergence. Specifically, $\omega$ depends on the spectral radius of the iteration matrix $\rho_{\text{jac}}$ as:

$$\omega = \frac{2}{1 + \sqrt{1 - \rho_{\text{jac}}^2}} \,. \tag{1}$$

The optimal $\omega$ for a full square grid can be approximated as:

$$\omega_{\text{opt}} \approx \frac{2}{1 + \frac{\pi}{N}} \,, \tag{2}$$

where $N$ is the number of grid points in one dimension. In our assignment we will use the following ansatz for the relaxation factor:

$$\omega = \frac{2}{1 + \alpha \frac{\pi}{N}} \,, \tag{3}$$

where $\alpha$ is a tunable parameter that we will need to optimize for our specific problem.

Since the SOR method iterative process is symmetric with respect to grid point parity, we can perform updates in a checkerboard pattern in two half-steps. This allows us to use an even faster approach for convergence called **Chebyshev Acceleration**, which is a technique that optimizes the relaxation factor $\omega$ over the course of half-iteration steps as:

$$\omega^{(}0) = 1\,, \tag{4}$$

$$\omega^{(1/2)} = \frac{1}{1 - \frac{1}{2}\rho_{\text{jac}}^2}\,, \tag{5}$$

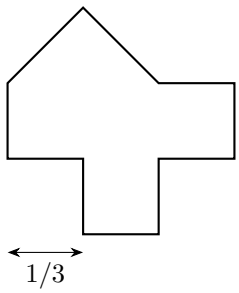
$$\omega^{(n)+1/2} = \frac{1}{1 - \frac{1}{4}\omega^{(n)}\rho_{\text{jac}}^2}\,. \tag{6}$$

This approach allows us to achieve convergence in a number of iterations that scales as $\mathcal{O}(N)$, which is a significant improvement over the $\mathcal{O}(N^2)$ scaling of the standard SOR method. For this to work however we need to have a good estimate of the spectral radius $\rho_{\text{jac}}$, which we can obtain by tuning $\alpha$ in Equation (3) and using the optimal $\omega$ in Equation (1) to get $\rho_{\text{jac}}$.

# 2 Task

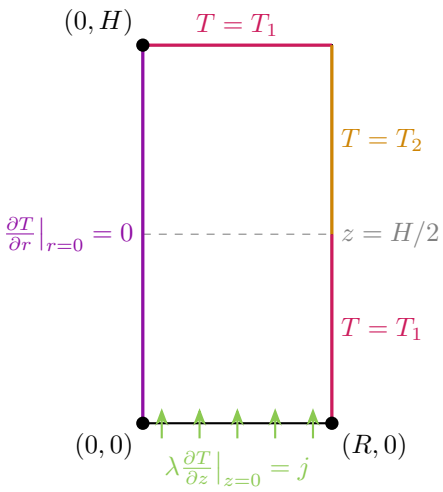
## 2.1 Steady-State Velocity Profile and Poisseuille Coefficient

Using the SOR method, we will solve the Poisson equation for the velocity profile of a fluid flowing through a pipe with a *'house-like'* cross-section. We are instructed to pay close attention to the correct treatment of cases where the boundary conditions are not defined at the grid points, but are instead defined at the midpoints between grid points. We need to find the optimal parameter $\alpha$ for the relaxation factor $\omega$ and test out the Chebyshev acceleration method. Finally, we need to compute the Poisseuille coefficient for the flow.

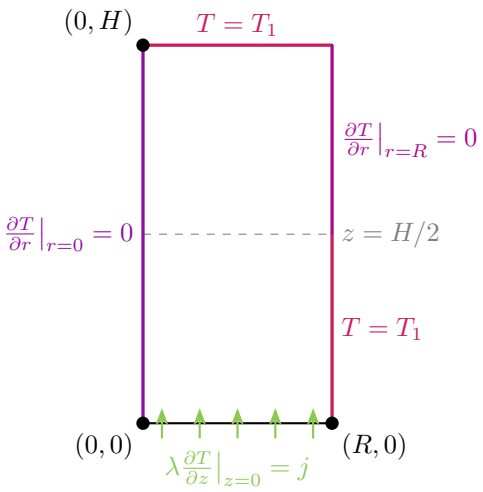



## 2.2 Steady-State Temperature Distribution in a Cylinder

Next, we will solve the steady-state heat equation in an equilateral cylindrical geometry with a constant heat flux at the bottom of the cylinder. The cylindrical surface is cut horizontally in half, with each half at a different temperature. The top surface of the cylinder is kept at the same temperature as the lower half of the cylinder. Our goal is to find the steady-state temperature distribution. We need to then repeat this calculation for the same geometry, but where the upper half of the cylinder is thermally insulated.




(a) Case 1: **Dirichlet** boundary condition at the upper half of the cylinder.




(b) Case 2: **Neumann** boundary condition at the upper half of the cylinder.

# 3 Solution Overview

Ha! This week was perhaps more involved than the previous ones. I guess you could say that I was up to my *old shenanigans* again. This week the standard Python computational stack of `numpy`, `scipy`, and `matplotlib` was augmented by a favorite of mine `mpi4py`, which provides bindings for the Message Passing Interface (MPI) standard in Python.

As stated in the introduction, the SOR method is symmetric with respect to grid point parity, which means we can independently update the two bipartite sublattices of the grid in a checkerboard pattern. This makes parallelization of the SOR method via domain decomposition very straightforward. We can divide the full lattice into smaller subdomains, which I will refer to as "chunks", partially inspired by that one block game. Each chunk can be assigned to a different MPI rank, and the updates can be performed in parallel. The only communication that is required between the ranks is the exchange of boundary values between neighboring chunks after each half-step of the SOR iteration, but since we can independently update the two sublattices *ghost halo* broadcasting very simple and efficient.

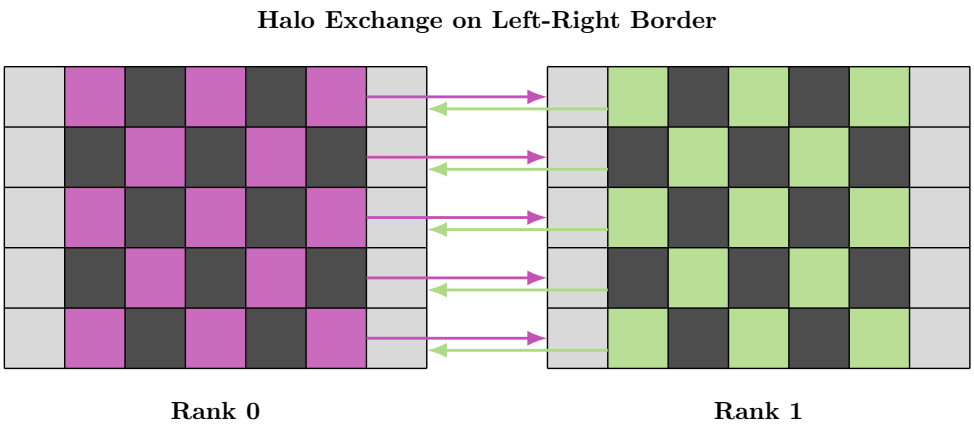



Figure 2: Ranks with Purple-Black and Green-Black sub-lattices exchanging a ghost column on their shared border.

I'll skip the various details of the implementation, otherwise I'll never beat the *too much comp-sci* allegations. But I will say that the implementation was very stable even with differing rank topologies and sublattice sizes. Overriding the base Cartesian class to implement the cylindrical Laplace was also quite simple. As for boundary point handling, I used the well known **Shortley-Weller** stencil [2], which is locally a first-order discretization, but globally second-order accurate [3]. The weights of the stencil are computed as:

$$w_R = \frac{1}{\theta_R(\theta_R + \theta_L)}\,, \tag{7}$$

$$w_L = \frac{1}{\theta_L(\theta_R + \theta_L)}\,, \tag{8}$$

$$w_U = \frac{1}{\theta_U(\theta_U + \theta_D)}\,, \tag{9}$$

$$w_D = \frac{1}{\theta_D(\theta_U + \theta_D)}\,, \tag{10}$$

where $\theta_R$, $\theta_L$, $\theta_U$, and $\theta_D$ are the fractional distances from the grid point to the boundary in the right, left, up, and down directions respectively. I computed fractional distances once at the beginning of the simulation by ray-casting from each boundary-adjacent point in each of the four directions to the boundary against the boundary contour that was extracted via the marching squares algorithm. The update at the boundary-adjacent interior point is then computed as a weighted average as:

$$\tilde{u}_i = \frac{w_R u_R + w_L u_L + w_U u_U + w_D u_D - q_i \Delta x^2}{w_R + w_L + w_U + w_D}\,, \tag{11}$$

such that the SOR update is then given by:

$$u_i^{(n+1)} = u_i^{(n)} + \omega \left( \tilde{u}_i - u_i^{(n)} \right) \tag{12}$$

I also found recent development in higher order stencils exactly for in solvers that use ghost halo exchange [4], however this is entirely unnecessary for our purposes. In fact, I found that the pure 5-point stencil (equivalent to Shortley-Weller with $\theta = 1$) is already sufficiently accurate for our purposes of computing the Poisseuille coefficient on the house-shaped domain. At $N \approx 400$ the problematic areas (diagonals and 270° corner points) are few in comparison to the rest of the domain, and the error is localized to those areas. The local area error is then of order $\mathcal{O}(\Delta x)$, but the global error, an integral over the whole domain, is of order $\mathcal{O}(\Delta x^2)$, which likely got *eaten* by my generous tolerance of $10^{-6}$ for convergence.

Before moving on the main results we can attempt to verify that our implementation works properly by applying it to test cases for which we know the analytical solution. We can start with the *trivial* case of a square domain with Dirichlet boundary conditions. We know that the Poisseuille coefficient for a unit square domain is $C_{\text{square}} = 0.0351443$ [5, Section 3-3]. We can sweep over $\alpha$ to find the optimal $\omega$ for which the number of iterations required for convergence is minimized, and also observe how the Poisseuille coefficient changes in that process. Figure 3 shows the results of the scan over $\alpha$ for the square case.

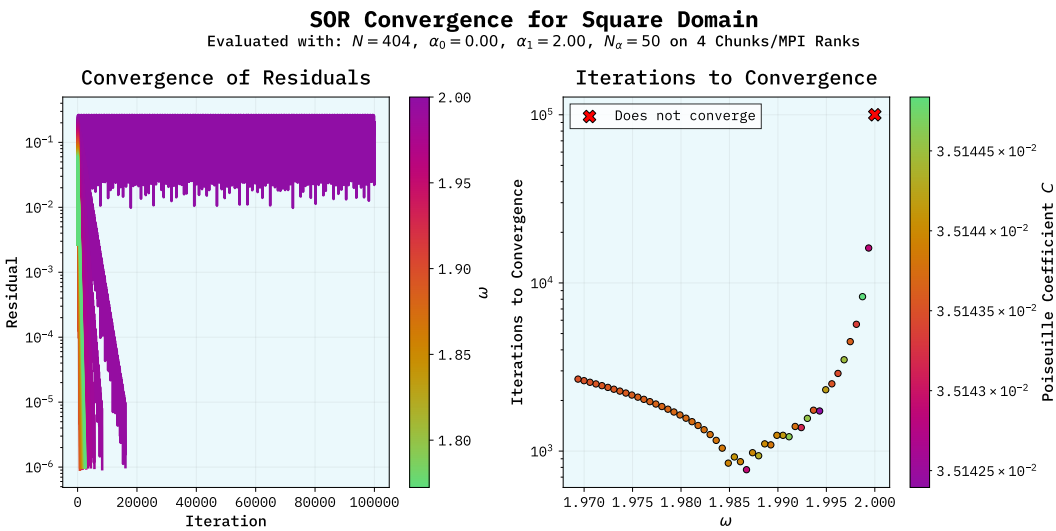



Figure 3: Scan over $\alpha$ for the square domain. The left-hand plot shows the convergence of the residuals for different values of $\alpha$, while the right-hand plot shows the number of iterations required for convergence and the computed Poisseuille coefficient for each value of $\alpha$.

I intentionally included the point $\alpha = 0.0$ in the scan, which corresponds to $\omega = 2.0$, to show that the method never converges, as is evident by the residuals not decreasing at all. Taking a more zoomed in look we see that the optimal value of $\omega$ is around 1.985, which for $N = 404$ corresponds to $\alpha \approx 0.98$, which is very close to the optimal value of $\alpha = 1.0$ that we would get from Equation (2). As seen from the colors on the right-hand scatter plot, we can see that the Poisseuille coefficient is very close to the analytical value for all values of $\alpha \in [0.0, 2.0]$, which were tested in this scan. So with this we've seen that our solver converges to the correct solution and that it does so at the optimal $\omega$ that is close to the theoretical prediction.

Next we can solve the Poisson equation for flow in a circular pipe. A classical result for the Poisseuille coefficient for a circular pipe is $C_{\text{circle}} = R^2/32 \approx 0.03125$ [6], where $R = 1$ is the radius of the pipe. Figure 4 shows the results of a wide scan over $\alpha$ for the circular pipe. It is clear that such wide scans are not very useful for finding the optimal $\alpha$ since they are too coarse, but they do show the general behavior of the convergence with respect to $\alpha$.

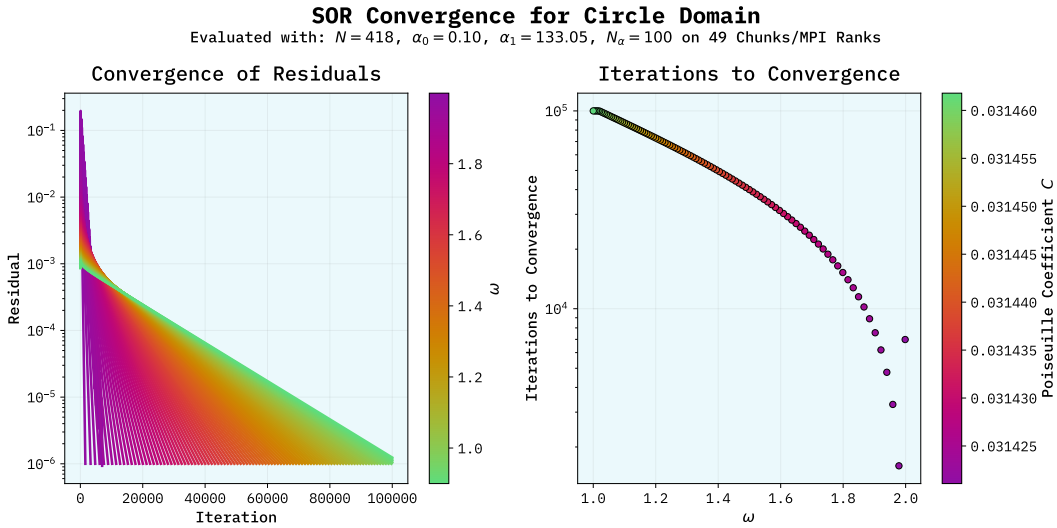



Figure 4

We can clearly see that there exists an optimal $\omega$ (and thus $\alpha$) for which the number of iterations required for convergence is minimized, however since our scan was wide and coarse we cannot determine the exact value from this plot. Figure 5 shows a zoomed in version of the scan around the optimal $\alpha$, where we can read that the optimal $\omega$ is around 1.985, which for $N = 418$ corresponds to $\alpha \approx 0.98$, which is the same optimal $\omega$ as for the square case, and is again very close to the optimal value of $\alpha = 1.0$ that we would get from Equation (2). This a sensible result if we consider that the optimal $\omega$ is determined by the spectral radius of the iteration matrix. The spectral radius is determined by the largest eigenvalue of the iteration matrix, which is tied to the *characteristic length* of the domain, **not** its shape. Since the characteristic length of the circular pipe is its radius, which is the same as the side length of the square domain, it makes sense that the optimal $\omega$ is the same for both cases.

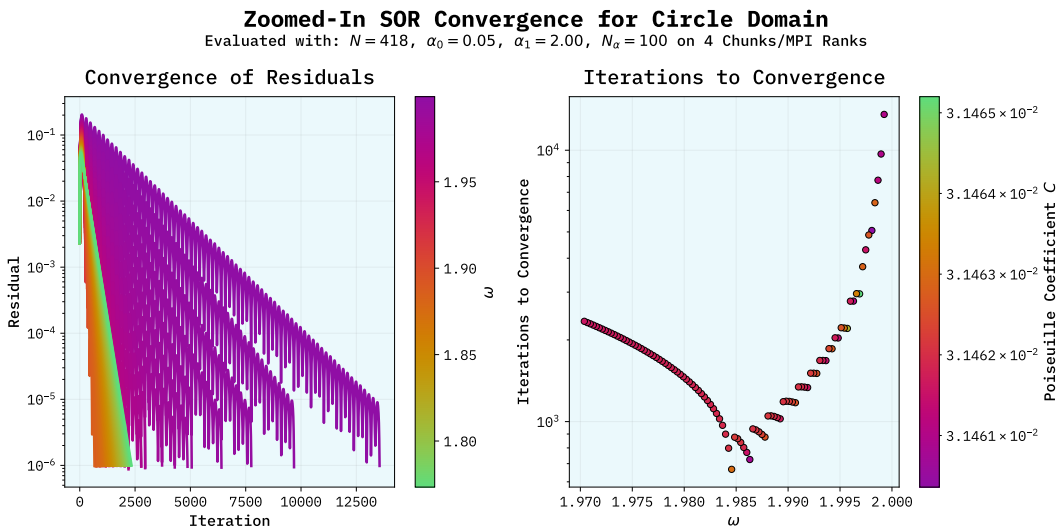



Figure 5

As for the Poisseuille coefficient, we can see on the wider scan that the values do change with $\alpha$, but they are all close regardless of the value of $\alpha$. Upon closer inspection for $\alpha \in [0.05, 2]$ we can see that the computed Poisseuille coefficient is very close to the analytical value, reaching $C_{\text{circle}}^{(\text{num})} = 0.03146$ at the optimal $\alpha$. With that I hope we've built a compelling case for the correctness of our implementation, and we can now move on to the main task of computing the velocity profile and Poisseuille coefficient for the house-shaped domain.

# 4 Results

Figure 6 shows the domain masks and the steady-state velocity profiles for the square, circular, and house-shaped domains. The velocity profiles are normalized by the maximum velocity in the domain, such that they are easier to compare.

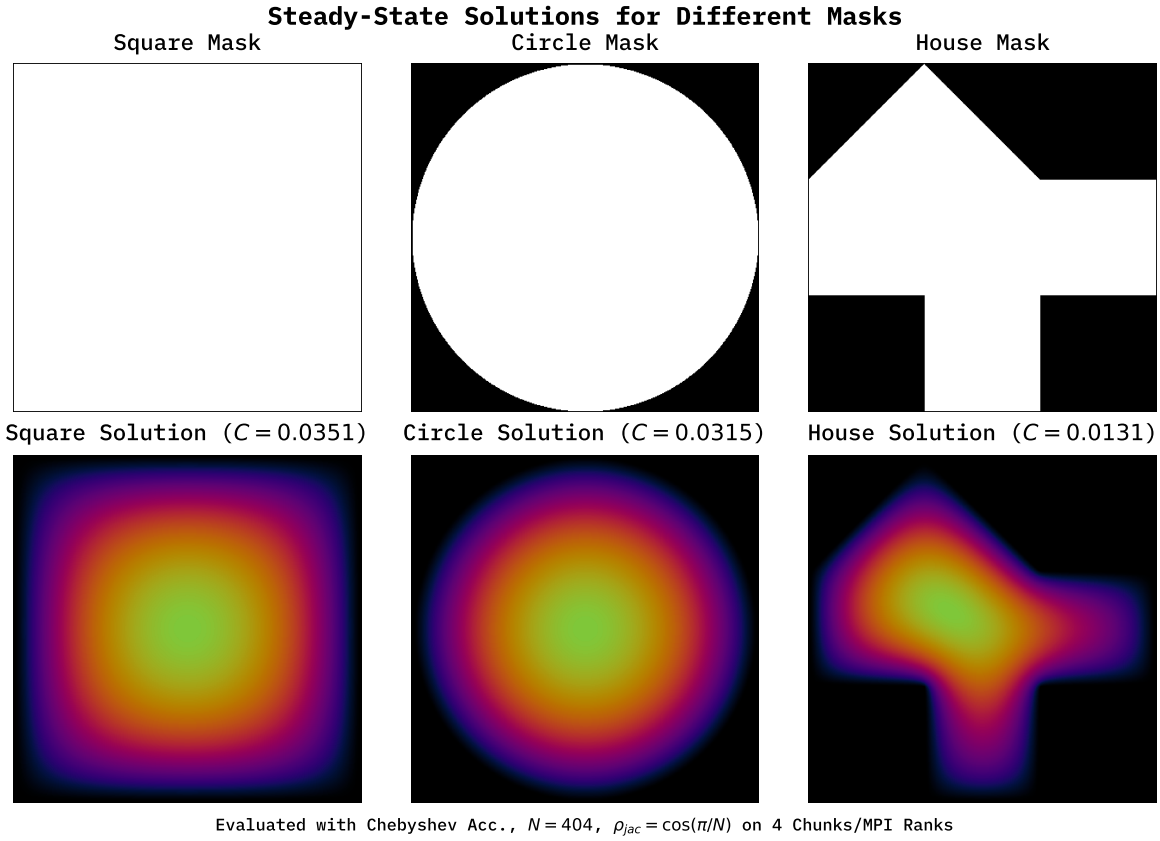



Figure 6: Domain masks and steady-state velocity profiles for the square, circular, and house-shaped domains. The velocity profiles are normalized by the maximum velocity in the domain.

The results are exactly as we expected them to be. The velocity profile for the square domain is the Fourier series solution, which essentially gives a smooth bell-shaped profile that flattens out near the boundaries, which we can see we have successfully reproduced with our solver. The velocity profile for the circular domain is the classic *Hagen-Poiseuille* profile, which is also parabolic, but with a different curvature. The velocity profile for the house-shaped domain is more complex, but it still has shapes that seem sensible. The widest part of the domain, where the base of the house intersects with the outcrop of a *'plus'* shape, has the highest velocity, while the narrowest parts of the domain, the corners of said outcrops, have the lowest velocity. This plot was computed using Chebyshev acceleration where the square approximation for the optimal $\omega$ was used to compute the spectral radius $\rho_{\text{jac}}$. We will more thoroughly examine the effect of Chebyshev acceleration shortly.

Figure 13 shows the wide and coarse scan over $\alpha$ for the house-shaped domain. Since this view is less useful for determining the optimal $\alpha$ it has been included in the Appendix, while the zoomed in version in Figure 7 is shown here.

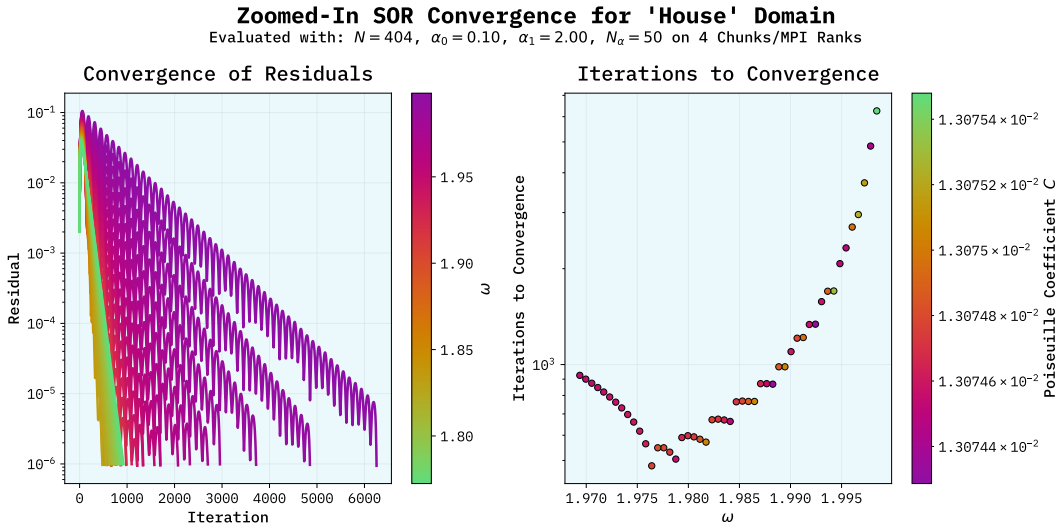



Figure 7: Zoomed in scan over $\alpha$ for the house-shaped domain. The left-hand plot shows the convergence of the residuals for different values of $\alpha$, while the right-hand plot shows the number of iterations required for convergence and the computed Poisseuille coefficient for each value of $\alpha$.

We can read from the right-hand plot that the optimal $\omega$ is around 1.977, which for $N = 404$ corresponds to $\alpha \approx 1.5$, which is somewhat higher than the square and circular cases, but is also sensible if we consider that the house-shaped domain has a larger characteristic length along the diagonal. The computed Poisseuille coefficient at the optimal $\alpha$ is $C_{\text{house}}^{(\text{num})} = 0.0131$. The computed Poisseuille coefficient is sensibly lower than the square and circular cases, which is expected since the house-shaped domain has a more complex geometry with narrow outcrops that should lead to higher resistance to flow.

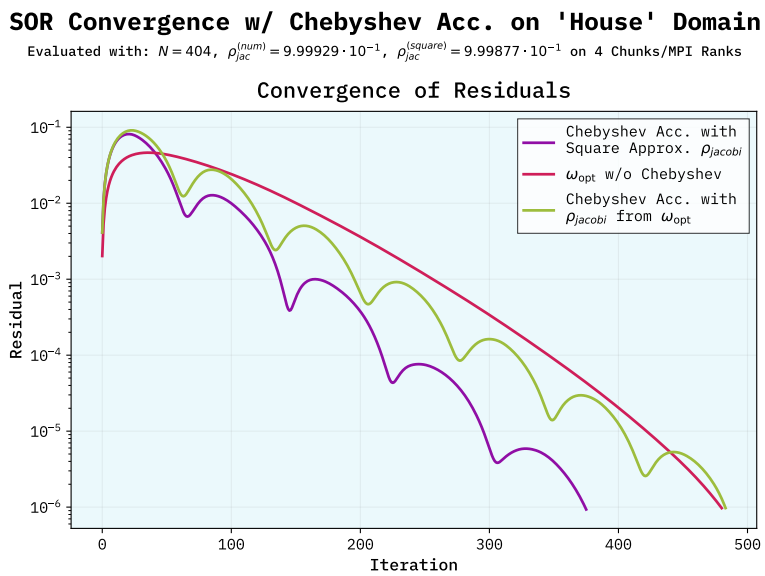



Figure 8: Comparison of the convergence of the residuals for the house-shaped domain without Chebyshev acceleration, with Chebyshev acceleration using the square approximation for $\rho_{\text{jac}}$, and with Chebyshev acceleration using the optimal $\rho_{\text{jac}}$ obtained from the optimal $\alpha$ via Equation (1).

Something I neglected to comment on earlier are the sequences of residuals for values of $\alpha$ that approach the stability limit of $\omega = 2.0$. Figure 3 showed clearly that we can expect divergent behavior for $\alpha = 0.0$ (i.e. $\omega = 2.0$) however, Figure 7 allows us to better see how residuals start to oscillate and take longer and longer to properly converge as we approach the stability limit. This is a well known behavior of the SOR method, and it is one of the reasons why it is important to find the optimal $\omega$ for convergence. I'm sure a more detailed analysis of the spectral radius of the iteration matrix could give us a more precise

estimate of the optimal $\omega$ for the house-shaped domain, but I think that the results we have are already quite good, and they are in line with our expectations based on the theoretical predictions.

Lastly for this subsection, Figure 8 shows the convergence of the residuals for the house-shaped domain without Chebyshev acceleration, with Chebyshev acceleration using the square approximation for $\rho_{\text{jac}}$, and with Chebyshev acceleration using the optimal $\rho_{\text{jac}}$ obtained from the optimal $\alpha$ via Equation (1). Oddly enough, the Chebyshev acceleration with the optimal $\rho_{\text{jac}}$ seems to perform worse than the one with the square approximation for $\rho_{\text{jac}}$. Even more oddly is the fact that the Chebyshev acceleration with the optimal $\rho_{\text{jac}}$ seems to perform worse than the standard SOR method without Chebyshev acceleration but at the optimal $\omega$. When thinking about this result, we can remind ourselves that the Chebyshev acceleration is designed to asymptotically approach the optimal $\omega$ in the course of the half-iteration steps, hence a more aggressive ($\rho_{\text{jac}}$ closer to 1) is the one that gets us closer to the optimal $\omega$ faster, which is likely why the square approximation for $\rho_{\text{jac}}$ performs better than the numerically obtained optimal $\rho_{\text{jac}}$. So my guess is that the slight underestimation of $\rho_{\text{jac}}$ by the square approximation is actually beneficial and produces a more aggressive Chebyshev acceleration that gets us closer to the optimal $\omega$ faster.

## 4.1 Steady-State Temperature Distribution in a Cylinder

The results for the steady-state temperature distribution in the cylinder are shown in Figure 9. The left-hand plot shows the temperature distribution for the case with Dirichlet boundary conditions at the upper half of the cylinder, while the right-hand plot shows the temperature distribution for the case with Neumann boundary conditions at the upper half of the cylinder. The results are exactly as we would expect them to be really, which is of course reassuring to see.

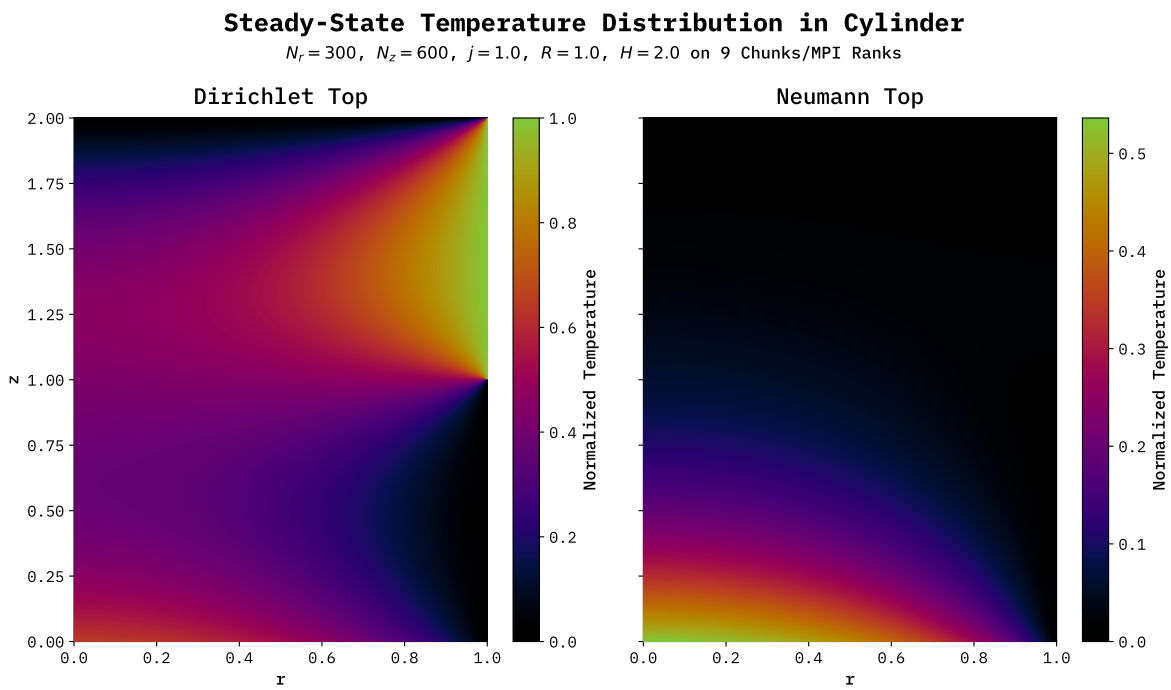



Figure 9: Steady-state temperature distribution in the cylinder for the case with Dirichlet boundary conditions at the upper half of the cylinder (left) and for the case with Neumann boundary conditions at the upper half of the cylinder (right).

I don't think I have too much to say about these results, since they are quite straightforward, since and they match our expectations based on the boundary conditions. I'm happy with the results and even more happy that I was able to generalize the SOR method to cylindrical coordinates while retaining all MPI parallelization features.
Figure 10 and Figure 11 show the convergence of the residuals for the two cases. We can see that the change of boundary conditions, while meaningfully changing the solution, does not change the optimal $\omega$ for convergence, which is again sensible since the optimal $\omega$ is from the domain not the boundary conditions. Other than that, no undiscussed behavior is observed here. The plots for iterations required

for convergence are very neat though. The optimal value sits again around $\alpha \approx 1.5$, which given the rectangular shape of the domain is also sane.

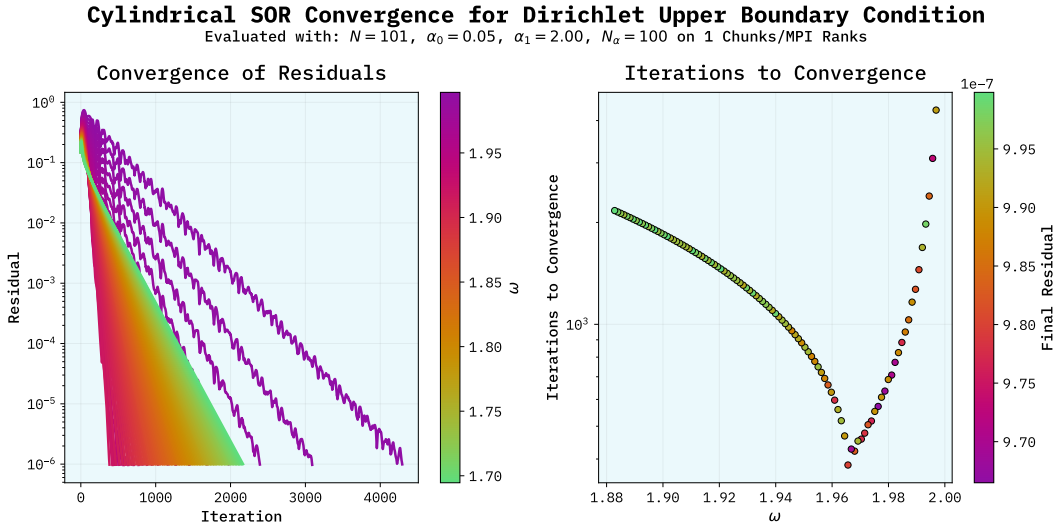



Figure 10: Convergence of the residuals for the case with Dirichlet boundary conditions at the upper half of the cylinder.

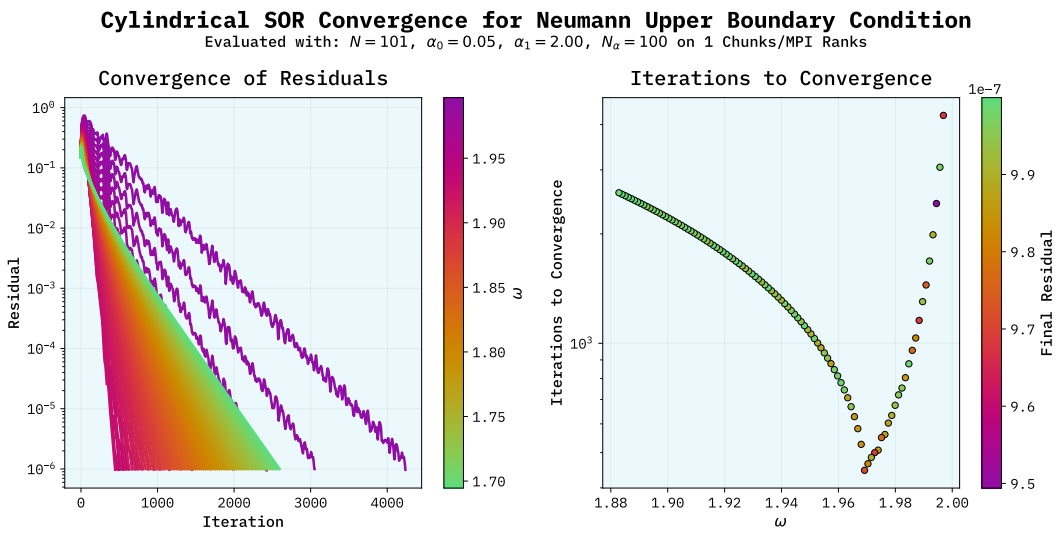



Figure 11: Convergence of the residuals for the case with Neumann boundary conditions at the upper half of the cylinder.

# 5 Conclusion and Comments

During the course of this report I think I was very scatter-brained, due to external *'life'* factors. I hope I still did a decent job of visualization and explanation of the results. I also hope that my mentions of technical details are not too overwhelming while not being too sparse. Something that would perhaps warrant a study or a plot would be how discretization errors affect the computed Poisseuille coefficient. I'm sure some of my colleagues will have plots to that tune. Since what we're doing is essentially a finite difference adjacent method we can expect the error to be of order $\mathcal{O}(\Delta x^2)$, but it would be nice to see that in practice. In my case I assumed that I had sufficiently high resolution for the error to be negligible, as also confirmed by the fact that the computed Poisseuille coefficients for the square and circular domains are very close to the analytical values.

To conclude, I want to add an additional pipe cross-section to the report that did not seem to fit well elsewhere. I want to prepend a disclaimer that this plot is ***in no way meant to be offensive*** to anyone, most of all our advisor/mentor whose likeness I used as the additional domain cross-section. The results for convergence and steady-flow velocity profile for the *'mentor'*-shaped domain are shown in Figure 12.

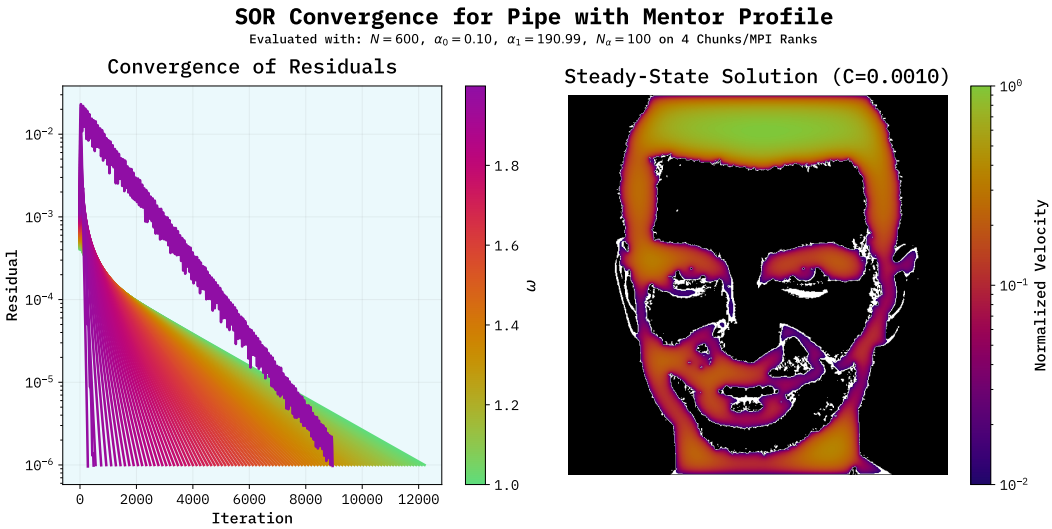



Figure 12: Convergence and steady-flow velocity profile for the *'mentor'*-shaped domain.

I think this is a fascinating result and would warrant way more discussion especially for other shapes. We see the velocity profile seems sensible, with the highest velocity in the widest part of the domain. The solver also does a decent job of converging to the solution even in very narrow areas of the domain, which is quite impressive. There definitely is room for improvement in very narrow areas, like the ears, but I think the results are already quite good. Besides a fun addition to the report, I wanted to include this plot to showcase the robustness and generality of the MPI parallelized SOR solver that I conjured up, which seems to be happy to handle arbitrary shapes without any issues.

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# A Additional Plots

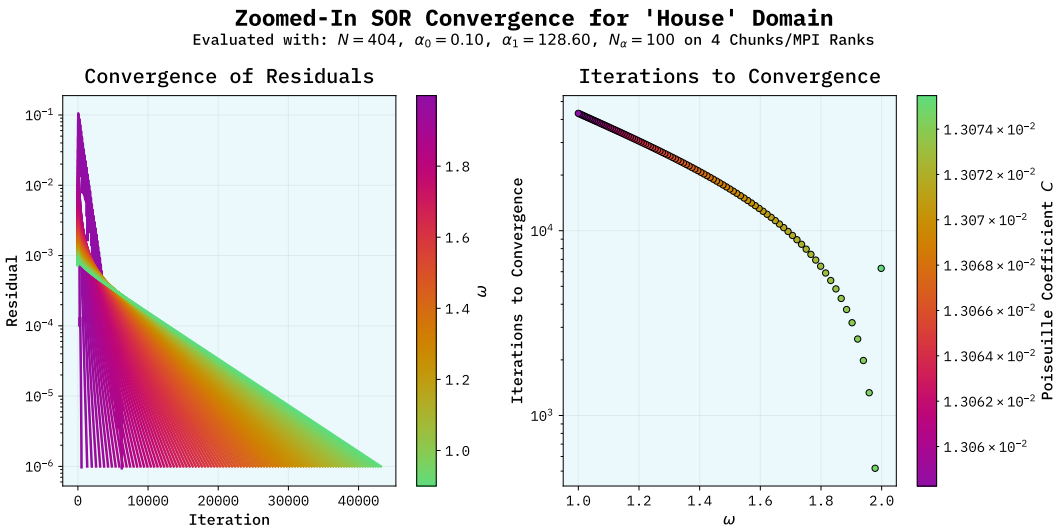


Figure 13: Wide and coarse scan over $\alpha$ for the house-shaped domain. The left-hand plot shows the convergence of the residuals for different values of $\alpha$, while the right-hand plot shows the number of iterations required for convergence and the computed Poisseuille coefficient for each value of $\alpha$. Plot primarily included for completeness, since it is not very useful for determining the optimal $\alpha$.